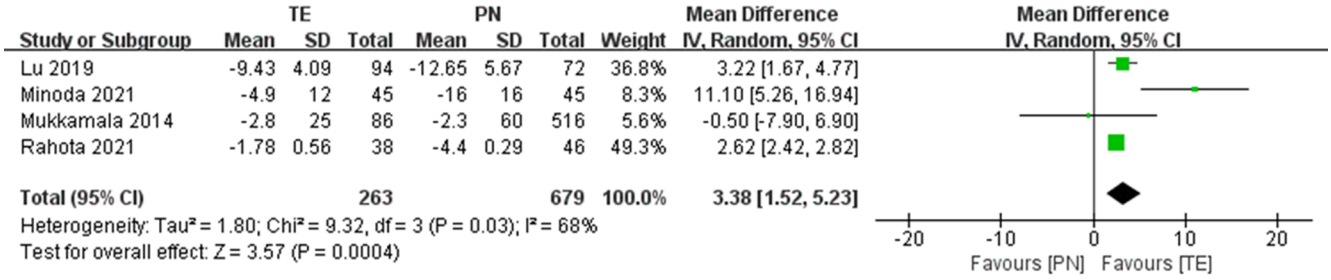
Supplementary Fig. Forest plot.

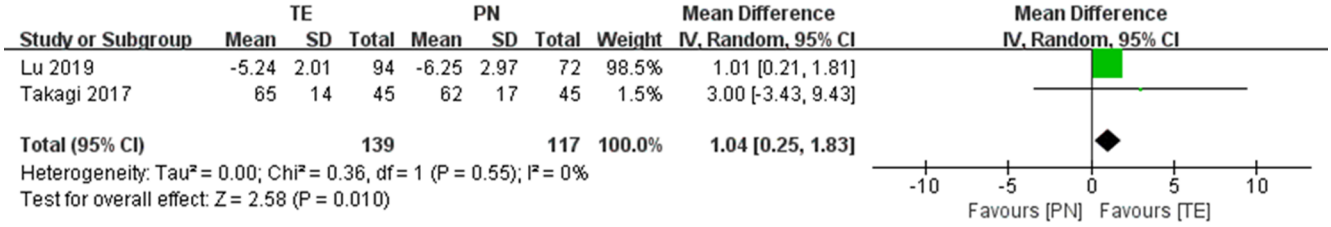
Primary outcomes

1. Residual renal function

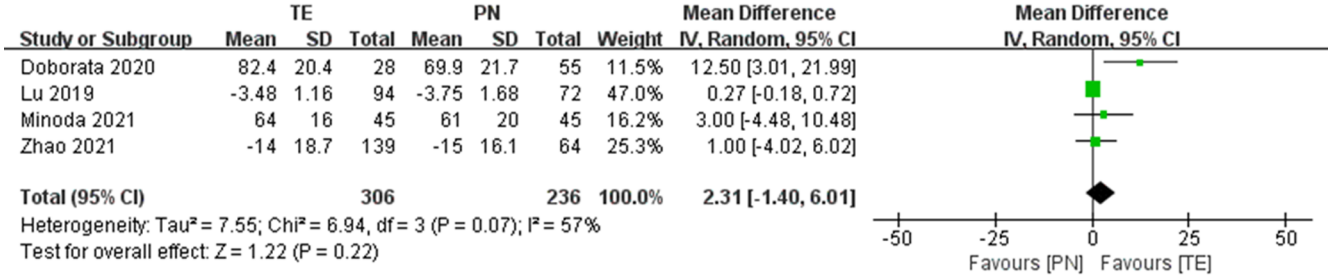
1.1. Perioperative (within 1 week)



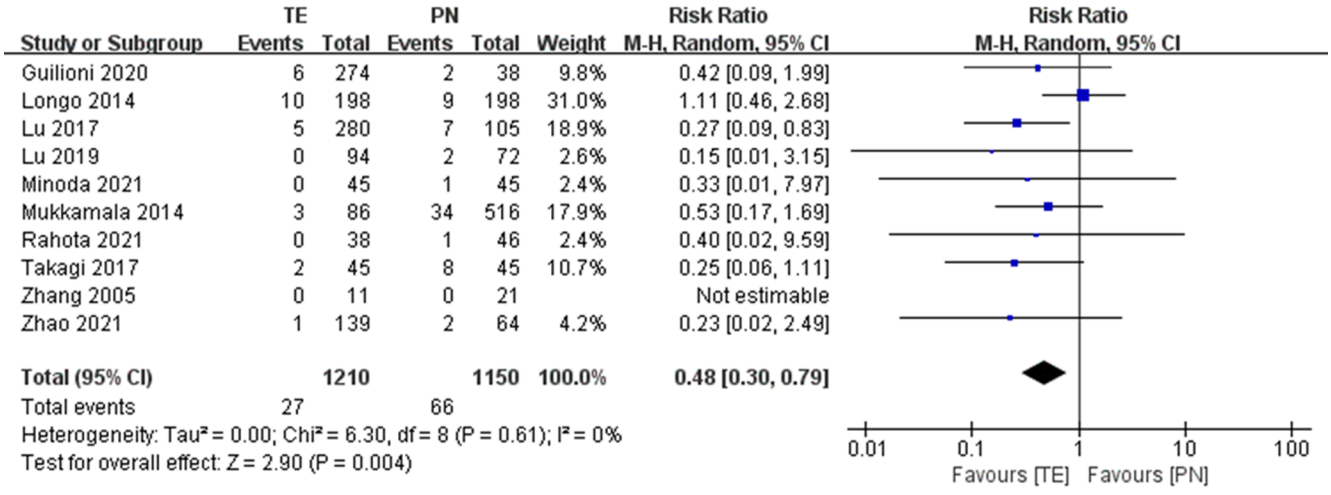
1.2. Short-term (up to 3 months)



1.3. Long-term (beyond 3 months)

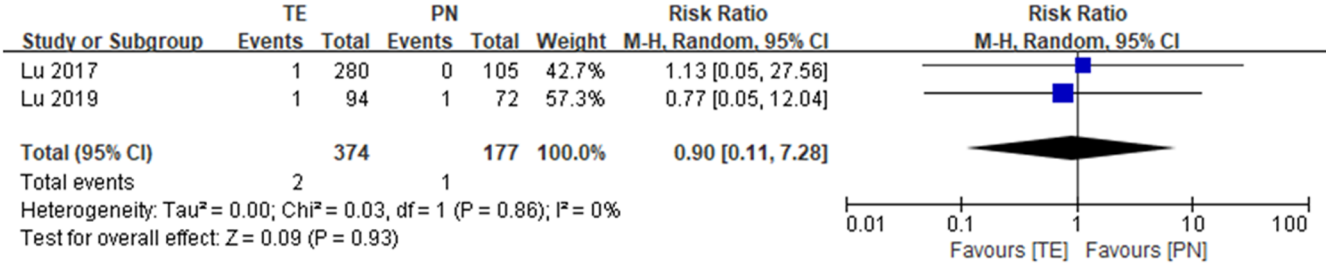


3. Major adverse events



Secondary outcomes

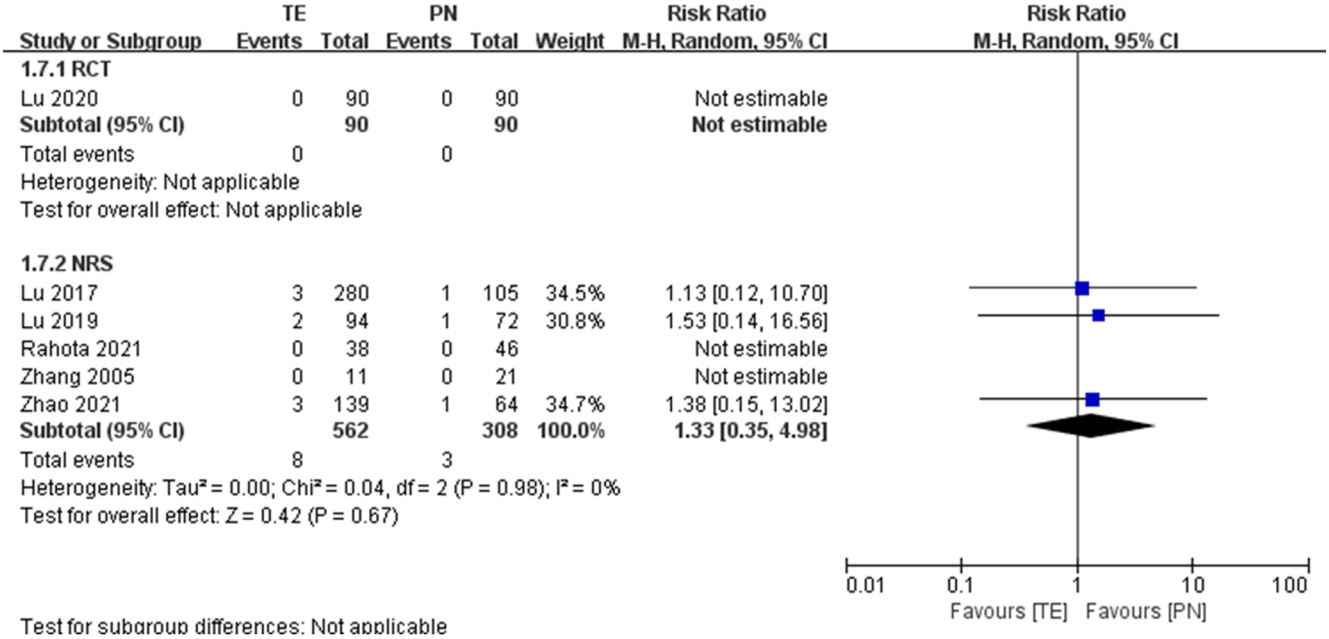
1. Cancer specific mortality



Study or Subgroup	TE Events	TE Total	PN Events	PN Total	Weight	Risk Ratio M-H, Random, 95% CI
Lu 2017	1	280	0	105	42.7%	1.13 [0.05, 27.56]
Lu 2019	1	94	1	72	57.3%	0.77 [0.05, 12.04]
Total (95% CI)		**374**		**177**	**100.0%**	**0.90 [0.11, 7.28]**
Total events	2		1			

Heterogeneity: Tau² = 0.00; Chi² = 0.03, df = 1 (P = 0.86); I² = 0%
Test for overall effect: Z = 0.09 (P = 0.93)

2. Local recurrence



Study or Subgroup	TE Events	TE Total	PN Events	PN Total	Weight	Risk Ratio M-H, Random, 95% CI
1.7.1 RCT						
Lu 2020	0	90	0	90		Not estimable
Subtotal (95% CI)		**90**		**90**		**Not estimable**
Total events	0		0			

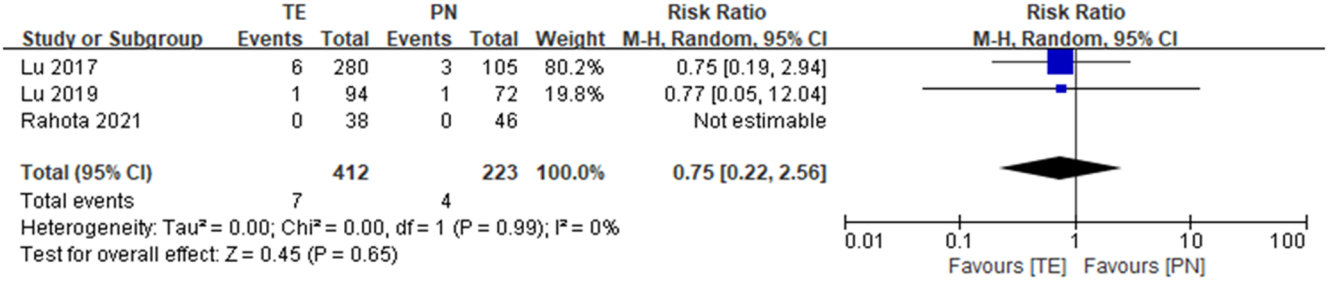
Heterogeneity: Not applicable
Test for overall effect: Not applicable

1.7.2 NRS						
Lu 2017	3	280	1	105	34.5%	1.13 [0.12, 10.70]
Lu 2019	2	94	1	72	30.8%	1.53 [0.14, 16.56]
Rahota 2021	0	38	0	46		Not estimable
Zhang 2005	0	11	0	21		Not estimable
Zhao 2021	3	139	1	64	34.7%	1.38 [0.15, 13.02]
Subtotal (95% CI)		**562**		**308**	**100.0%**	**1.33 [0.35, 4.98]**
Total events	8		3			

Heterogeneity: Tau² = 0.00; Chi² = 0.04, df = 2 (P = 0.98); I² = 0%
Test for overall effect: Z = 0.42 (P = 0.67)

Test for subgroup differences: Not applicable

3. Distant metastasis

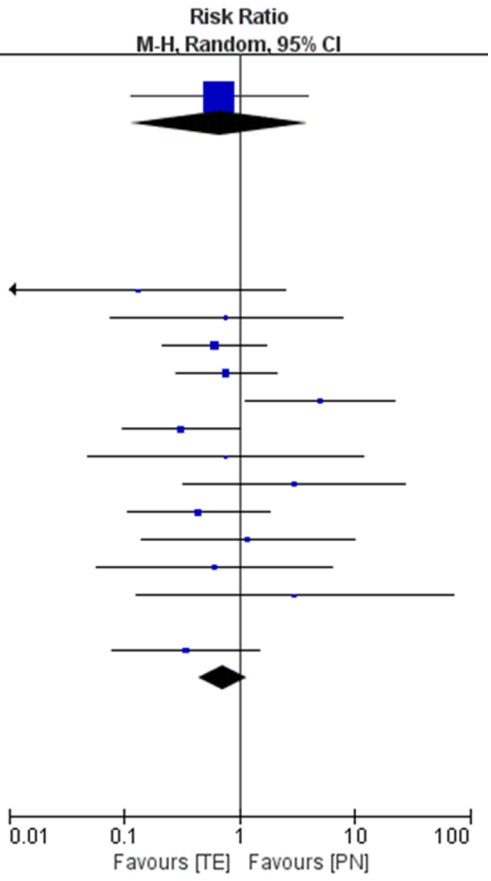


Study or Subgroup	TE Events	TE Total	PN Events	PN Total	Weight	Risk Ratio M-H, Random, 95% CI
Lu 2017	6	280	3	105	80.2%	0.75 [0.19, 2.94]
Lu 2019	1	94	1	72	19.8%	0.77 [0.05, 12.04]
Rahota 2021	0	38	0	46		Not estimable
Total (95% CI)		**412**		**223**	**100.0%**	**0.75 [0.22, 2.56]**
Total events	7		4			

Heterogeneity: Tau² = 0.00; Chi² = 0.00, df = 1 (P = 0.99); I² = 0%
Test for overall effect: Z = 0.45 (P = 0.65)

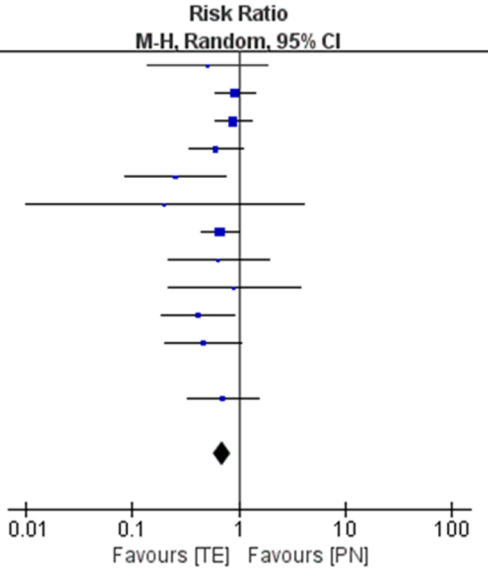
4. Positive surgical margin

Study or Subgroup	TE Events	TE Total	PN Events	PN Total	Weight	Risk Ratio M-H, Random, 95% CI
1.9.1 RCT						
Lu 2020	2	90	3	90	100.0%	0.67 [0.11, 3.90]
Subtotal (95% CI)		**90**		**90**	**100.0%**	**0.67 [0.11, 3.90]**
Total events	2		3			
Heterogeneity: Not applicable						
Test for overall effect: Z = 0.45 (P = 0.65)						
1.9.2 NRS						
Blackwell 2017	0	57	3	53	2.6%	0.13 [0.01, 2.52]
Calaway 2017	2	34	1	13	4.1%	0.76 [0.08, 7.73]
Gayarre 2020	5	26	6	19	16.3%	0.61 [0.22, 1.70]
Guilioni 2020	22	274	4	38	16.7%	0.76 [0.28, 2.09]
Longo 2014	10	198	2	198	8.9%	5.00 [1.11, 22.53]
Lu 2017	5	280	6	105	13.5%	0.31 [0.10, 1.00]
Lu 2019	1	94	1	72	3.0%	0.77 [0.05, 12.04]
Minoda 2021	3	45	1	45	4.5%	3.00 [0.32, 27.76]
Mukkamala 2014	2	68	26	391	9.9%	0.44 [0.11, 1.82]
Pogosyan 2020	1	19	4	90	4.8%	1.18 [0.14, 10.01]
Rahota 2021	1	38	2	46	4.0%	0.61 [0.06, 6.42]
Takagi 2017	1	45	0	45	2.3%	3.00 [0.13, 71.74]
Zhang 2005	0	11	0	21		Not estimable
Zhao 2021	3	139	4	64	9.3%	0.35 [0.08, 1.50]
Subtotal (95% CI)		**1328**		**1200**	**100.0%**	**0.72 [0.44, 1.17]**
Total events	56		60			
Heterogeneity: Tau² = 0.11; Chi² = 13.85, df = 12 (P = 0.31); I² = 13%						
Test for overall effect: Z = 1.32 (P = 0.19)						
Test for subgroup differences: Chi² = 0.01, df = 1 (P = 0.94), I² = 0%						



5. Overall adverse events

Study or Subgroup	TE Events	TE Total	PN Events	PN Total	Weight	Risk Ratio M-H, Random, 95% CI
Gayarre 2020	3	26	5	22	2.1%	0.51 [0.14, 1.89]
Guilioni 2020	94	274	14	38	18.0%	0.93 [0.60, 1.46]
Longo 2014	37	198	42	198	23.0%	0.88 [0.59, 1.31]
Lu 2017	26	280	16	105	10.7%	0.61 [0.34, 1.09]
Lu 2019	4	94	12	72	3.0%	0.26 [0.09, 0.76]
Minoda 2021	0	45	2	45	0.4%	0.20 [0.01, 4.05]
Mukkamala 2014	19	86	172	516	20.9%	0.66 [0.44, 1.00]
Pogosyan 2020	3	19	22	90	3.0%	0.65 [0.21, 1.94]
Rahota 2021	3	38	4	46	1.8%	0.91 [0.22, 3.81]
Takagi 2017	7	45	17	45	6.0%	0.41 [0.19, 0.90]
Wang 2017	7	59	15	58	5.3%	0.46 [0.20, 1.04]
Zhang 2005	0	11	0	21		Not estimable
Zhao 2021	14	139	9	64	5.9%	0.72 [0.33, 1.57]
Total (95% CI)		**1314**		**1320**	**100.0%**	**0.69 [0.57, 0.83]**
Total events	217		330			
Heterogeneity: Tau² = 0.00; Chi² = 10.38, df = 11 (P = 0.50); I² = 0%						
Test for overall effect: Z = 3.84 (P = 0.0001)						



6. Hospital stay

Study or Subgroup	TE Mean	TE SD	TE Total	PN Mean	PN SD	PN Total	Weight	Mean Difference IV, Random, 95% CI
Gayarre 2020	4.5	2.14	26	5.4	3.2	22	1.7%	-0.90 [-2.47, 0.67]
Lu 2017	8.3	3.1	280	8.7	3.2	105	8.1%	-0.40 [-1.11, 0.31]
Lu 2019	5.99	0.67	94	6.8	1.02	72	55.6%	-0.81 [-1.08, -0.54]
Minoda 2021	3.6	1.1	45	3.9	1.4	45	15.2%	-0.30 [-0.82, 0.22]
Takagi 2017	4.1	1.8	45	5	0.3	45	14.4%	-0.90 [-1.43, -0.37]
Zhao 2021	6.5	2.7	139	7.2	3.2	64	5.0%	-0.70 [-1.60, 0.20]
Total (95% CI)		**629**				**353**	**100.0%**	**-0.71 [-0.91, -0.51]**
Heterogeneity: Tau² = 0.00; Chi² = 4.18, df = 5 (P = 0.52); I² = 0%								
Test for overall effect: Z = 6.85 (P < 0.00001)								

